

- But physical systems are in their nature often **linear**, and excitations also tend to take sinusoidal shapes. $\Rightarrow$ Step-response analysis will not suffice and we need to analyze...

Engineering student classic: Tacoma-Narrows bridge

For this we introduce a new plot...

The Bode Plot

Quick frequency domain recap:

- Many signals can be written as a $e^{j\omega t}$ of $e^{-j\omega t}$:

$$u(t) =$$

Example: $u(t) = \cos(\omega t) =$

Applying $u(t)$ to an LTI-system $G(s)$, we get:

$$y_{ss}(t) =$$
 , so here we get

$$y_{ss}(t) =$$

=> hard to interpret because of complex part! Idea:

Portray complex value in $e^{j\omega t}$ form with

$e^{j\omega t}$, $e^{-j\omega t}$, with which we can write a complex number as $M e^{j\varphi}$. Applying this...

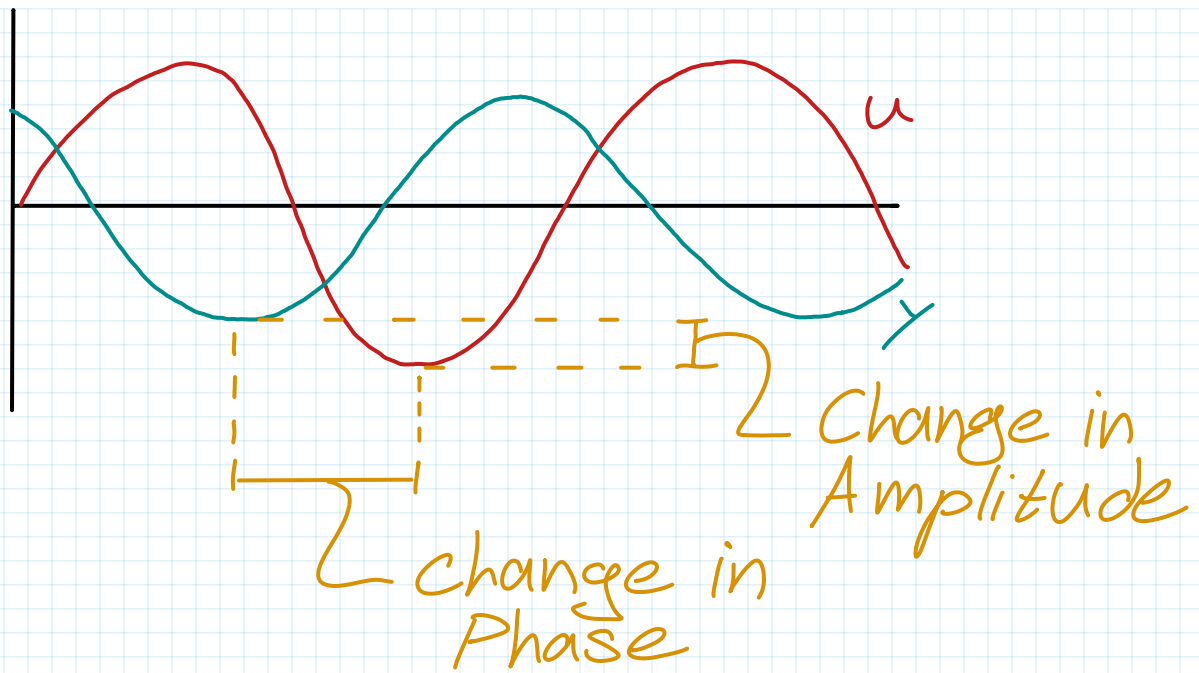
$$y_{ss} =$$

$$=$$

$$=$$

(input was $u(t) = \cos(\omega t)$)

=> M and φ depend on ω and fully characterize response!



• Important: For a given input with frequency ω , the steady-state response of an LTI-system has a constant

$$\cdot M(\omega) = |G(j\omega)|,$$

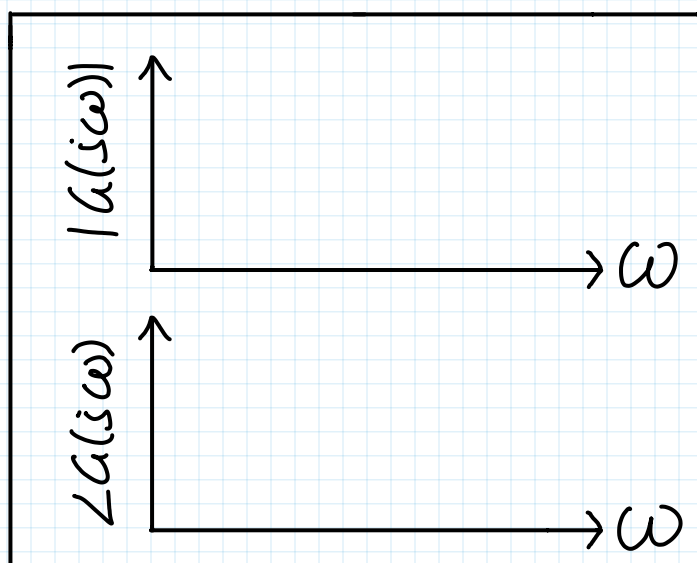
$$\cdot \varphi(\omega) = \angle G(j\omega),$$

and the frequency of the output is always also ω .

This is exactly what the Bode plot shows:
Amplitude and phase as a function of ω .

- The Bode plot therefore consists of two graphs as such:

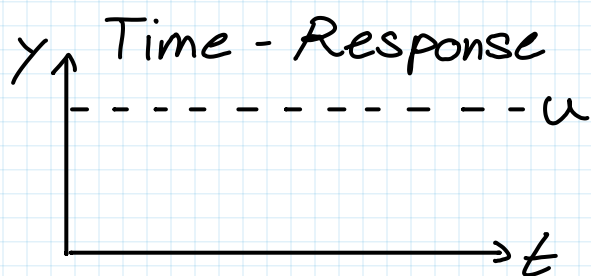
Bode Plot



- "Bode Plot": Both plots
- "Magnitude Plot"
- "Phase Plot"

What do you think Bode plots look like?

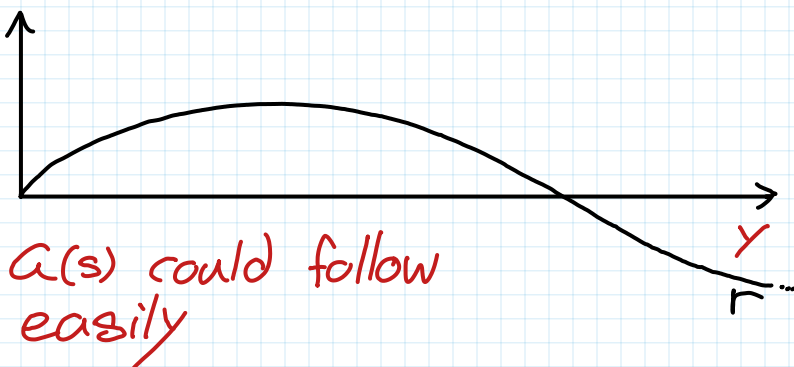
Building intuition:



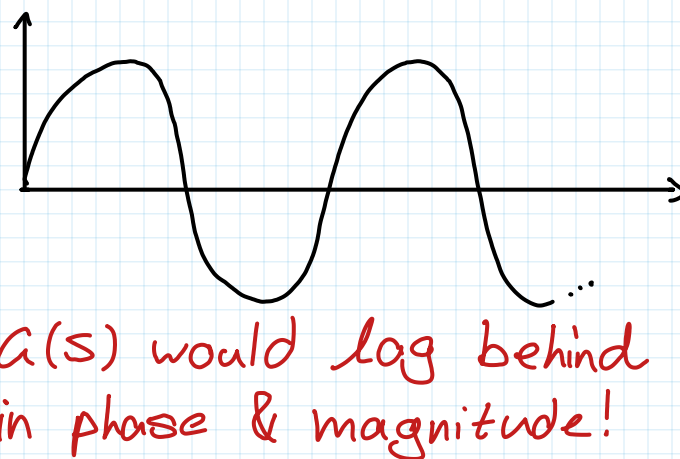
characterizes response of 1st. order system $G(s) =$

$\Rightarrow$ How fast is $G(s)$?

• Following low ω



• Following high ω



1) Scaling of vertical axis

$$|G(j\omega)| [dB] =$$

2) Scaling of horizontal axis

⇒ The log-log nature of the magnitude plot makes it hard to get intuition.

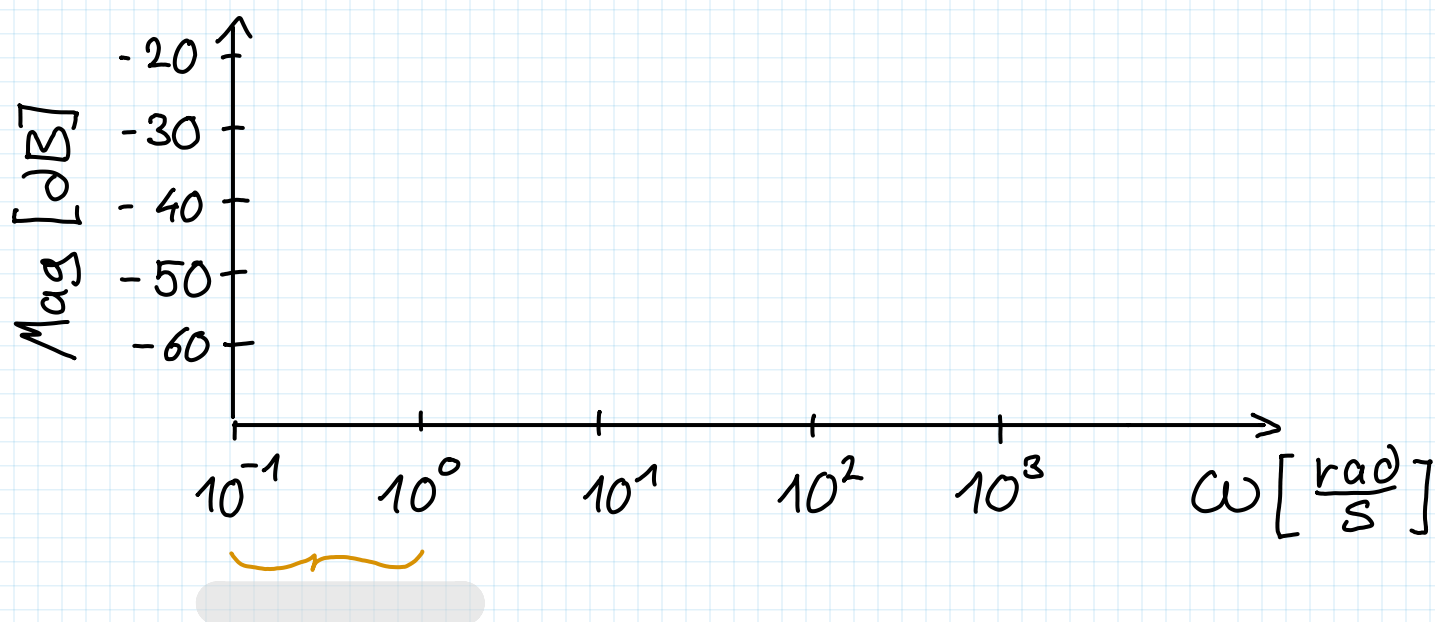
You are given the following system:

$$G(j\omega) = \frac{1}{s + 10} = \frac{1}{j\omega + 10}$$

• First we evaluate the magnitude plot:

$$|G(j\omega)| \text{ [dB]} =$$

$$= 20 \log_{10}(\quad). \quad \text{Plotting this:}$$



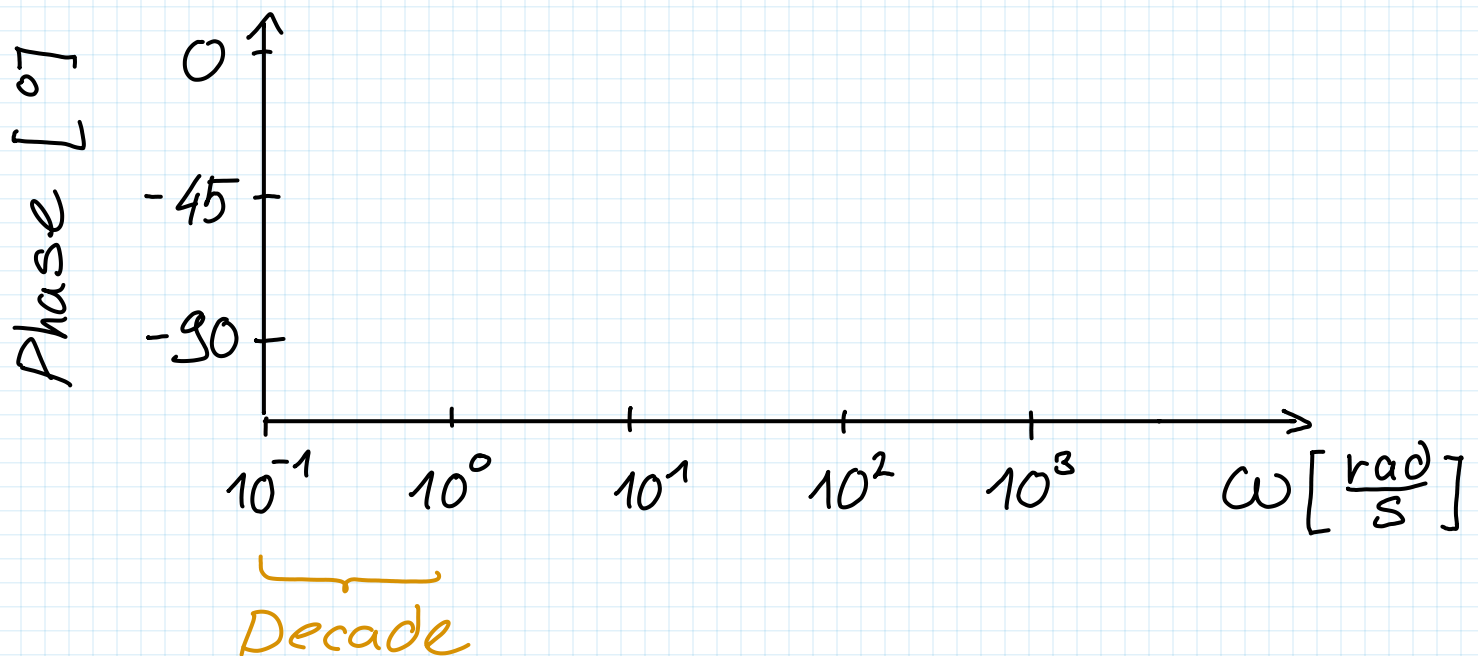
Before we discuss how we got to this, let's look at the phase plot:

• Evaluating the phase:

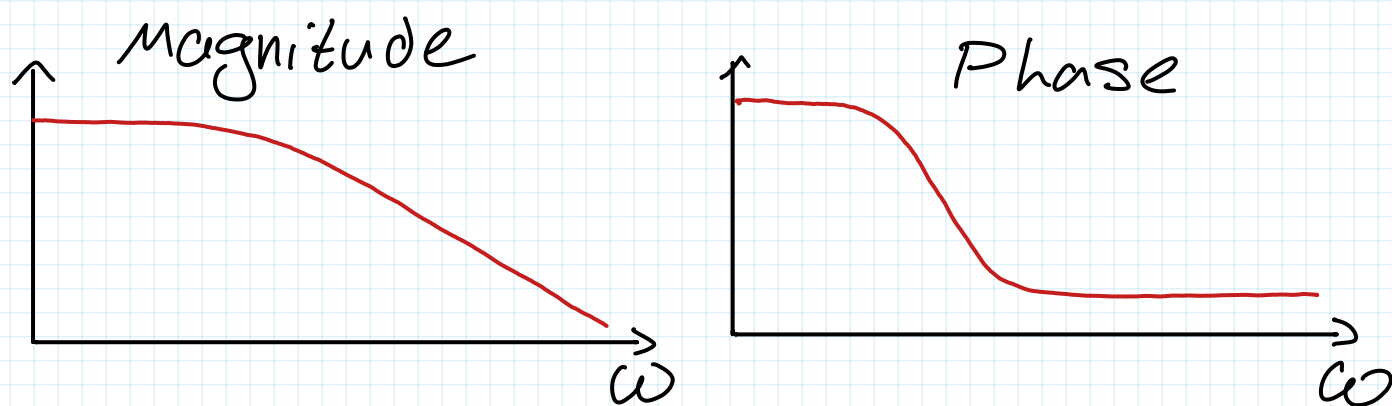
$$\angle G(j\omega) =$$

=

Plotting this:



So how can we interpret these plots?
What do they tell us about a system?



This is a very typical Bode configuration. Systems with this behaviour are also called

So how do we sketch them?

⇒ As usual, we analyze the poles and zeros:

Phase \ Magnitude	Magnitude	
	-20 dB/dec	+20 dB/dec
-90°	stable pole	non-minimum phase zero
+90°	unstable pole	minimum phase zero

They tell us how phase/gain changes over different frequencies.

Let's just try an example!

You are given a system $G(s)$

$$G(s) = \frac{(s+1)(s-100)}{(s+10)(s+30)(s-1000)}$$

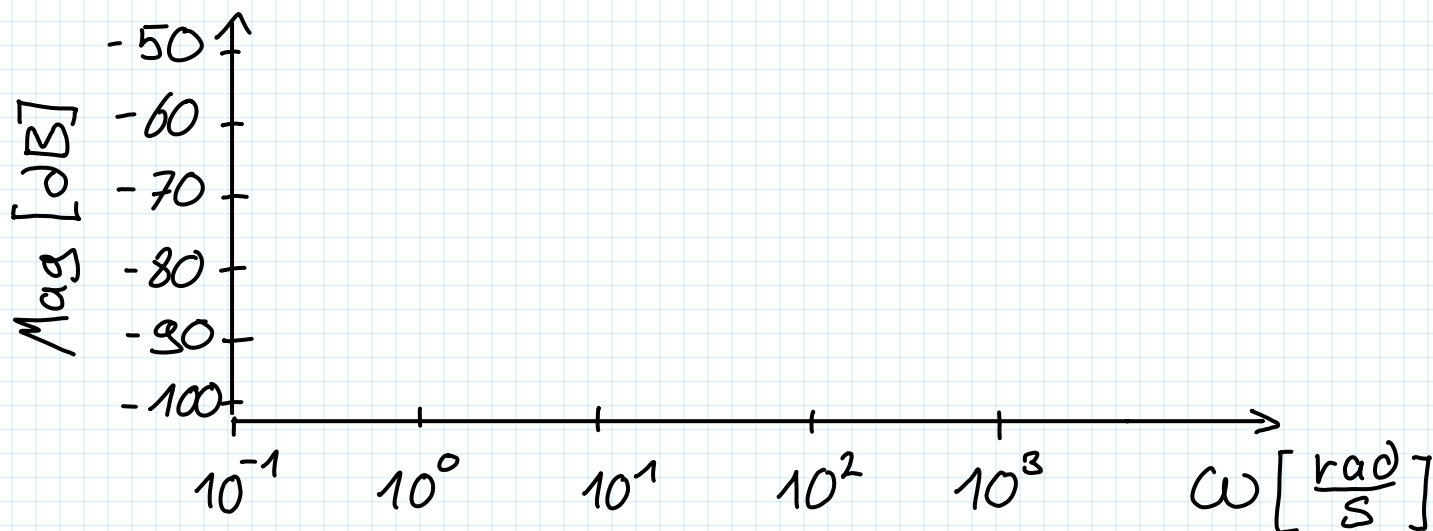
you would like to know how the steady-state frequency response looks like. $\Rightarrow$ Bode!

1) Find Bode-gain

$$G(0) =$$

$\Rightarrow$ Transform to [dB]

$$G(0)[\text{dB}] =$$



2) Sort poles & zeros from smallest to largest (absolute value):

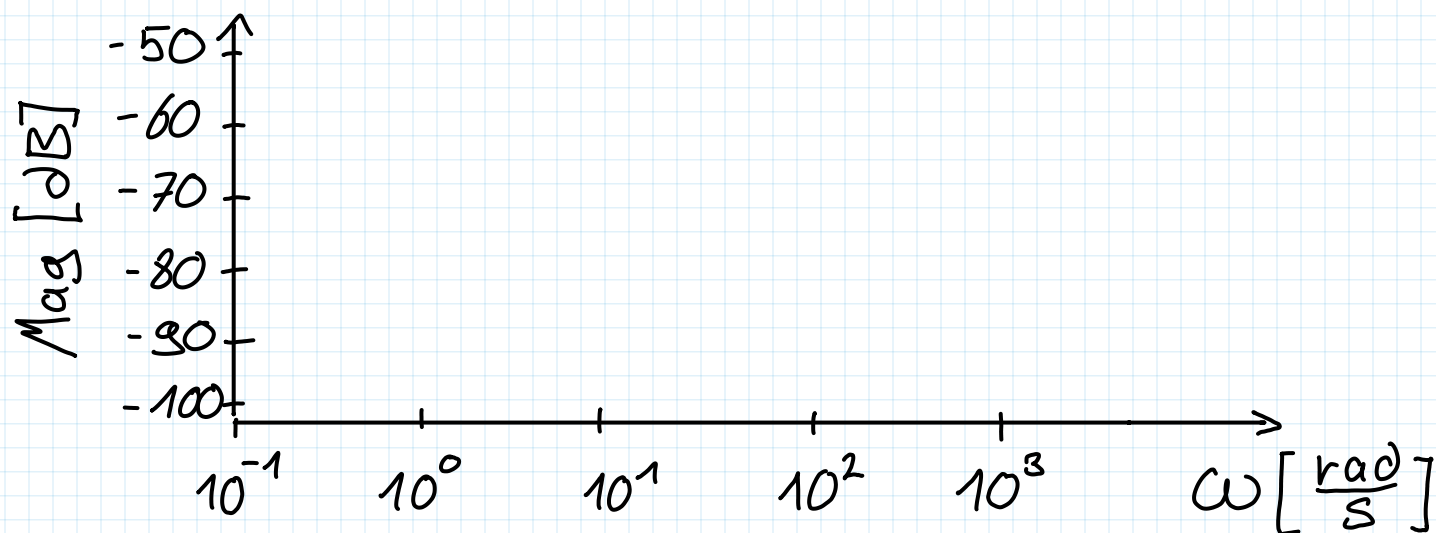
$$G(s) = \frac{(s+1)(s-100)}{(s+10)(s+30)(s-1000)}$$

smallest to largest (|·|)

$z_1 =$, $p_1 =$, $p_2 =$, $z_2 =$, $p_3 =$
 $z_1 =$, $p_1 =$, $p_2 =$, $z_2 =$, $p_3 =$

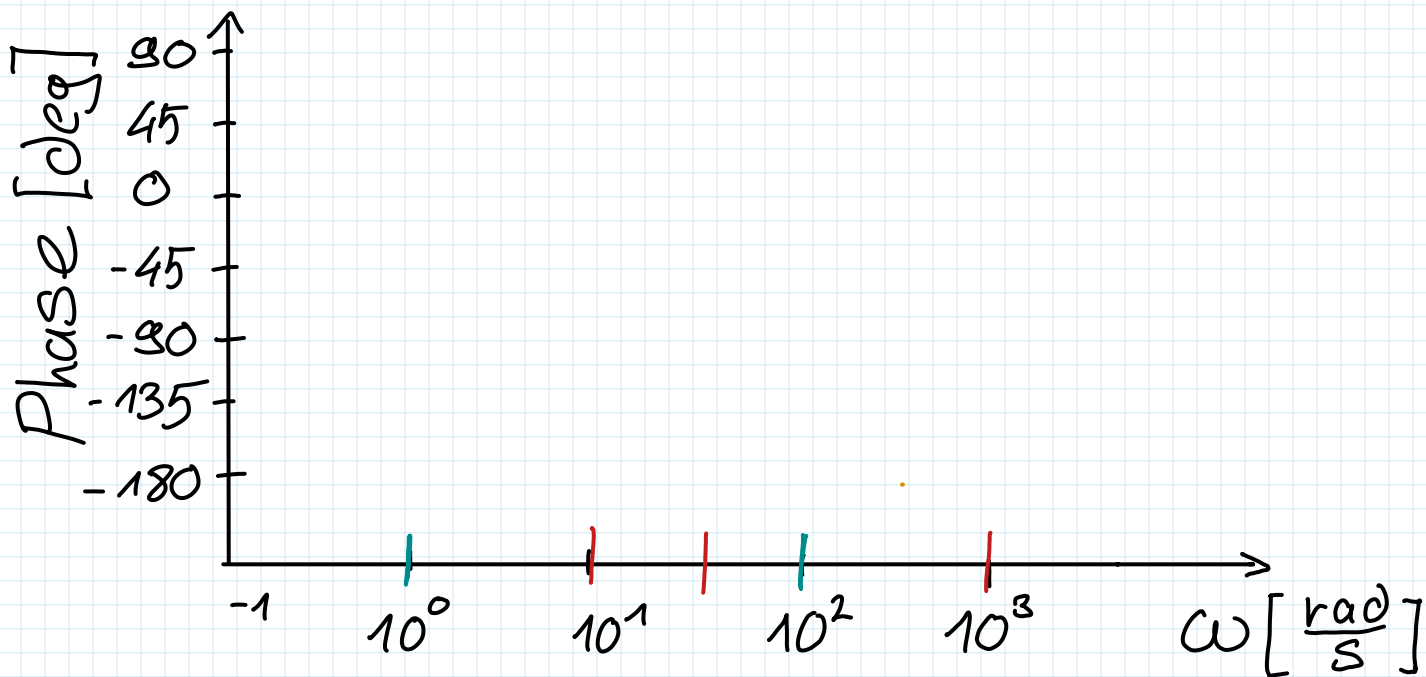
3) We follow the rules and change our steepness whenever we "encounter" a pole or zero

Phase	Magnitude	-20 dB/dec	+20 dB/dec
-90°		stable pole	non-minimum phase zero
+90°		unstable pole	minimum phase zero



Now for the phase plot:

We change the phase around a pole/zero.
The change is strongest at the pole/zero.



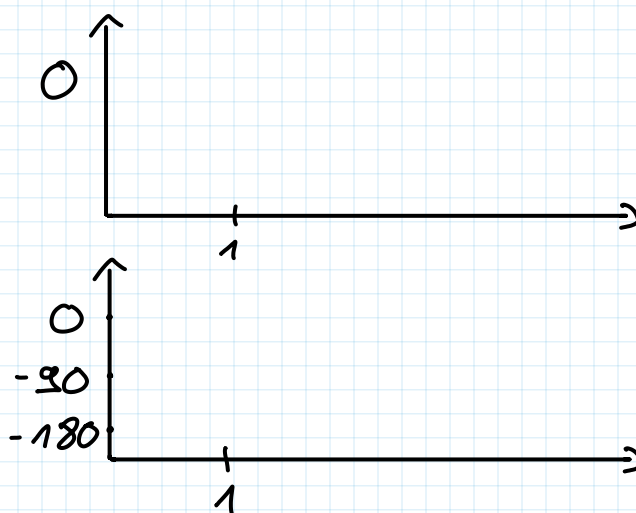
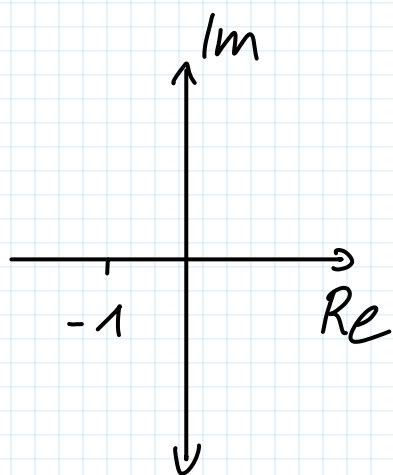
Special rules:

- If tf has pole $p=0$ (), the plot will start with and
- If tf has zero $z=0$ (), the plot will start with and

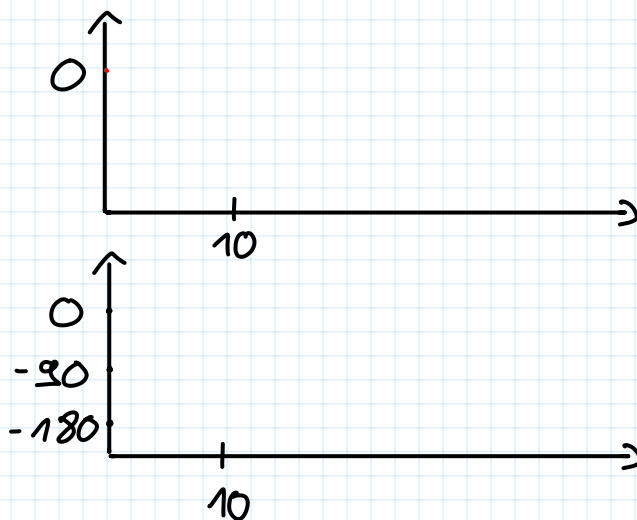
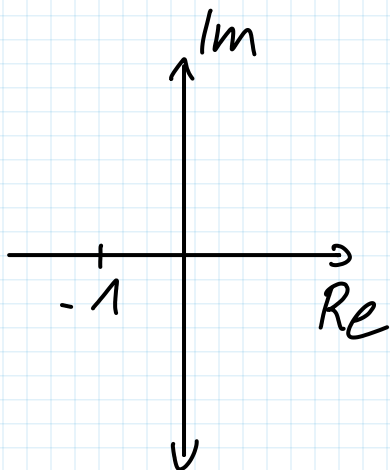
What if we have complex poles/zeros?

Let's compare the following systems:

$$\bullet G_1(s) = \frac{1}{(s+1)(s+1)}$$



$$\bullet G_2(s) = \frac{1}{(s+1+10j)(s+1-10j)}$$

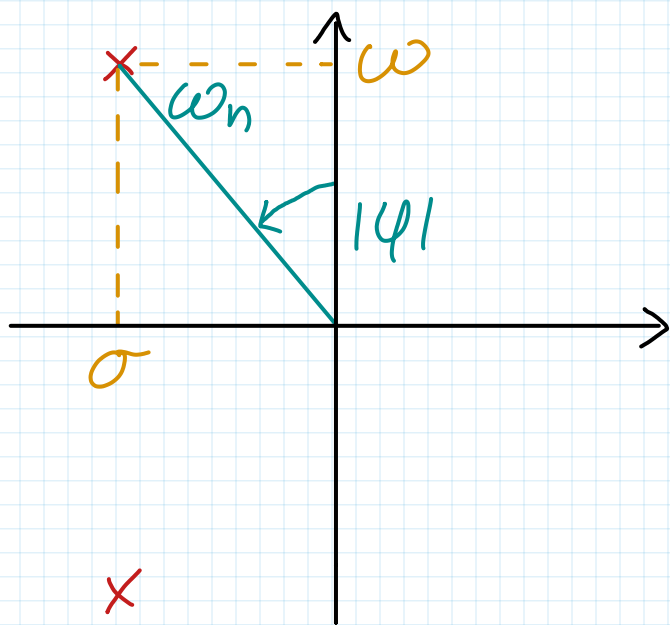


How can we characterize this peak?

• Remember how we wrote second order systems:

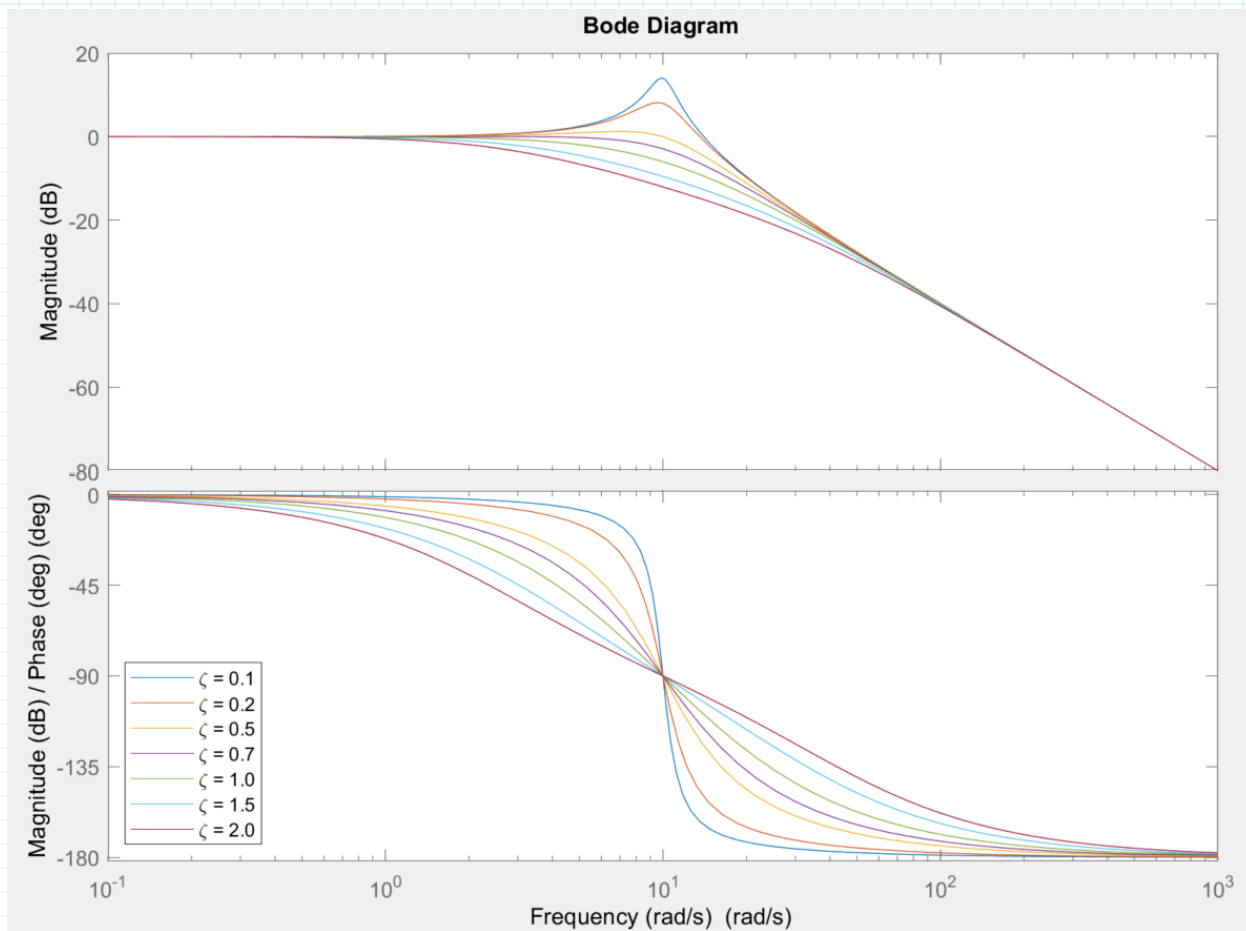
$$G(s) =$$

It turns out, that the damping factor ζ defines the of the resonance!



- $\omega_n = \sqrt{\sigma^2 + \omega^2}$
- $\varphi = \tan^{-1}\left(\frac{\sigma}{\omega}\right)$
- $|\sigma| = \sin(|\varphi|) \cdot \omega_n$
- $\sin(|\varphi|) = \zeta$

The stronger the damping, the smaller the resonance!



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

• Note: this is for systems, but the general principles stays the same for higher order systems.

• The Bode plot has one more great feature...

Given two systems, controller and plant:



• Magnitudes !

$$|L(s)| =$$

• Phases !

$$\angle L(s) =$$

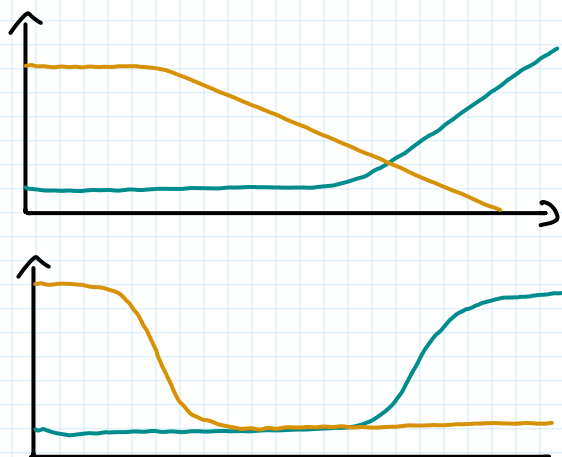
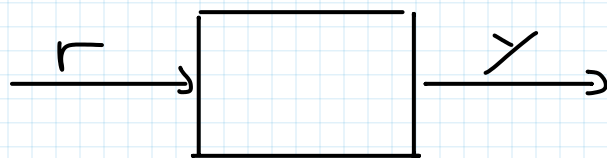
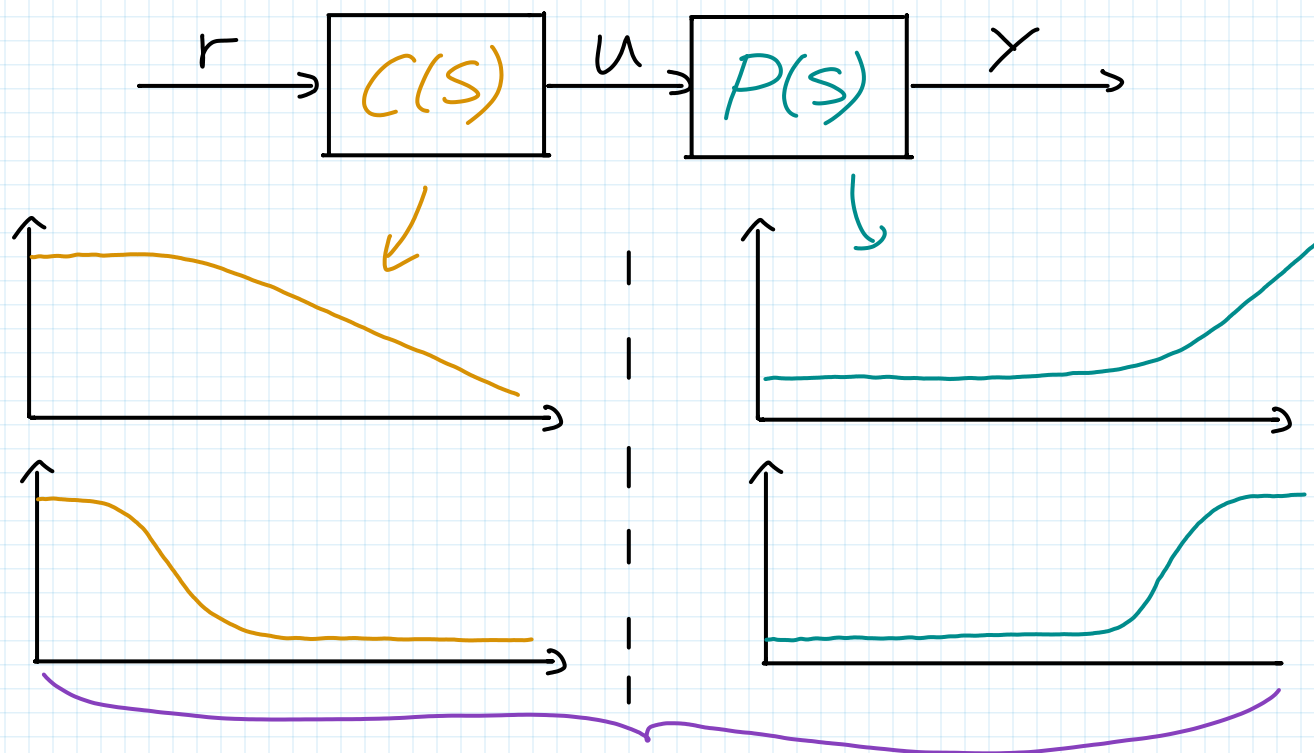
Remember logarithmic rules:

• The magnitude plot is logarithmic:

$$\log(|L(s)|) = \log(|C(s)| \cdot |P(s)|)$$

Thanks to this, we can just superpose Bode plots!

- If you have the Bode plot of $C(s)$ and $P(s)$, you can find the plot of $L(s)$ by adding them on top of each other!



- Thanks to this additive feature, we can also break down complicated tf's into simpler ones and recombine only the plots.

$$B(G(s)) \hat{=} \text{Bode plot of } G(s)$$

$$\cdot G(s)$$

$\Rightarrow$ Draw 1 plot

$$B(G)$$

$$\cdot G(s) =$$

$\Rightarrow$ Draw 3 plots

$$B(G_1), B(G_2), B(G_3)$$

$\Rightarrow$ Add them together

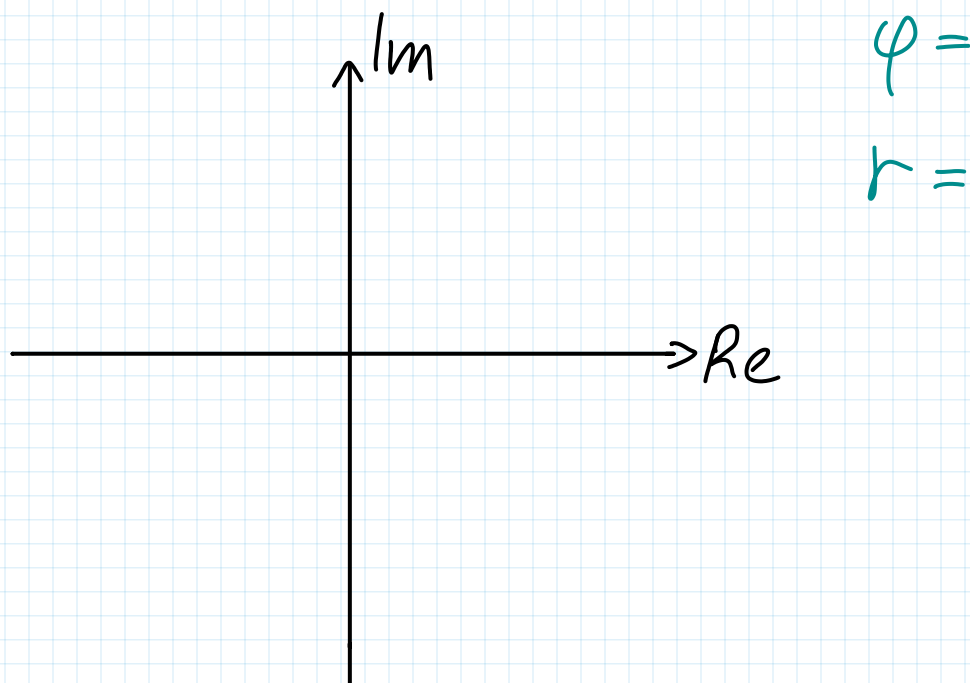
$$B(G) = B(G_1) + B(G_2) + B(G_3)$$

$\Rightarrow$ Same result, you can do it however you prefer.

What the Bode-plot represents can actually also be represented in just one plot...

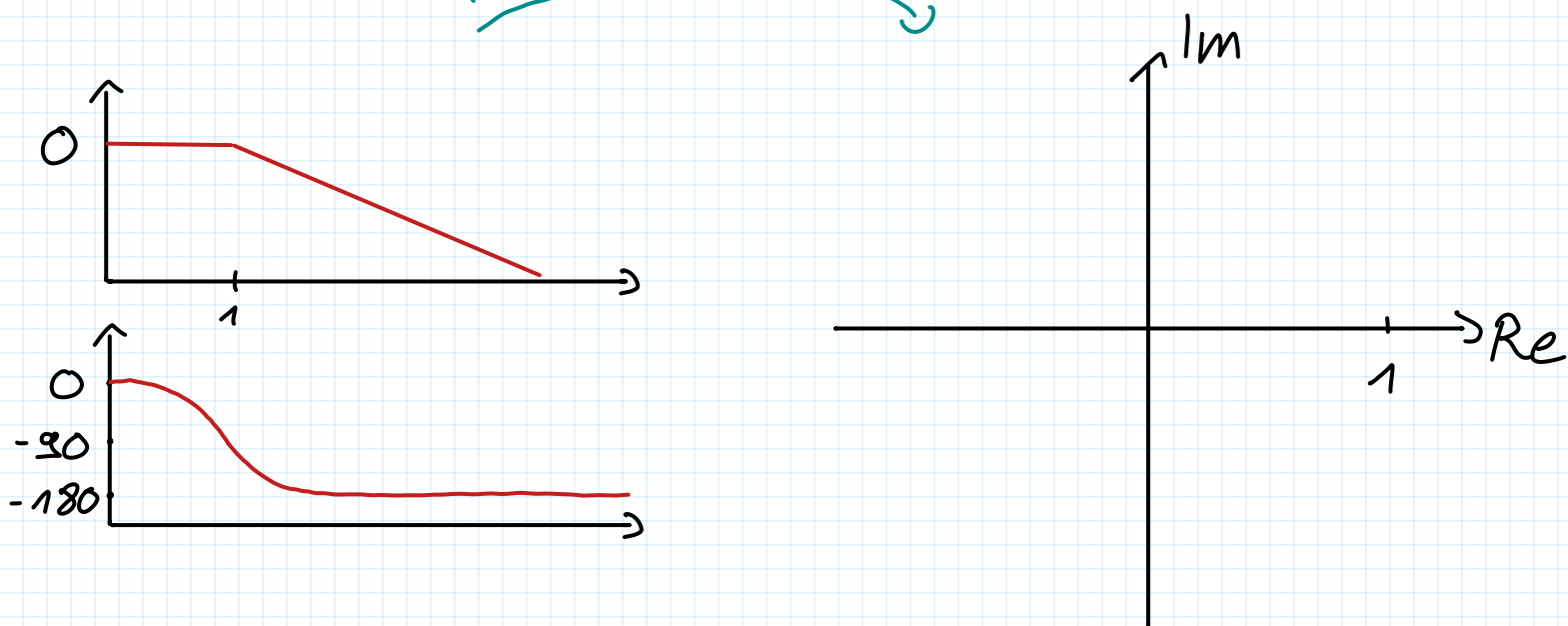
The polar plot

- In this plot we combine the magnitude and phase into a complex number .
- The frequency is no longer an axis, but ω , just like the gain k in the root locus



If we have the Bode plots, we extract the polar plot as such:

Bode to Polar



careful:

Note: For all of these plots we are sketching. you won't have to precisely draw them.

Ex: Find the Bode & Polar plots of $G(s)$

$$G(s) = \frac{(s-1)}{(s+10)(s+100)}$$

- First, identify poles/zeros

• Evaluate Bode gain:

$$G(0) =$$

$$\angle(G(0)) = \quad (\text{Phase plot starts at } 180^\circ)$$

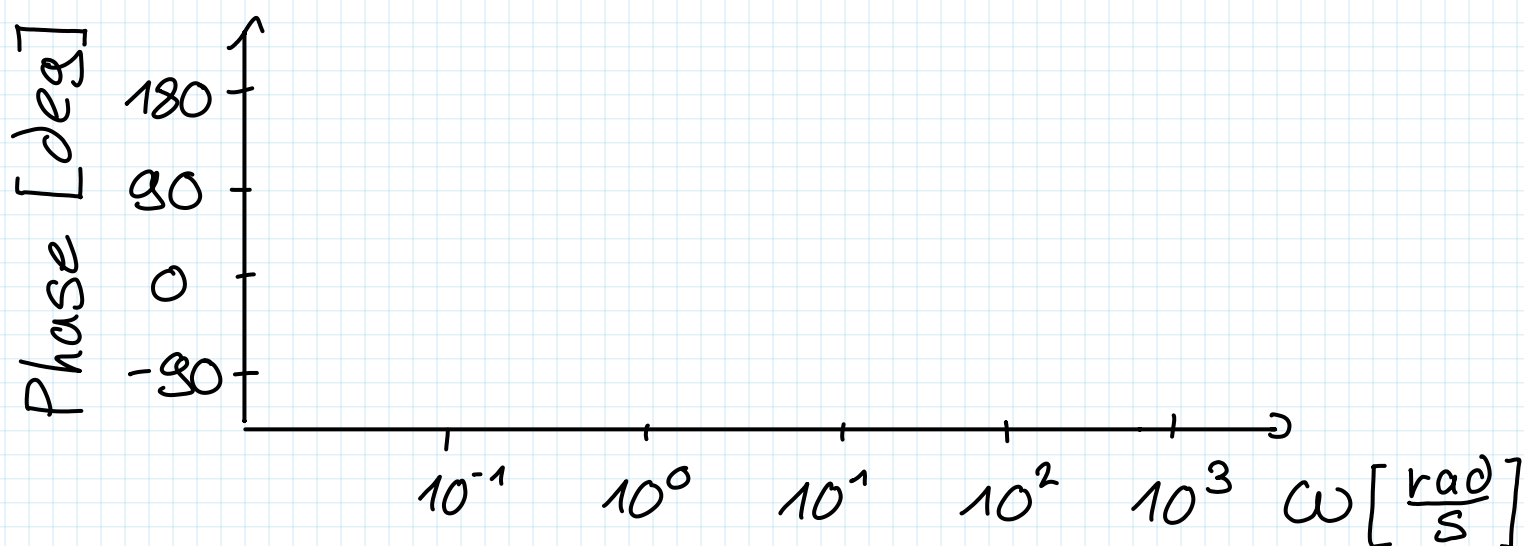
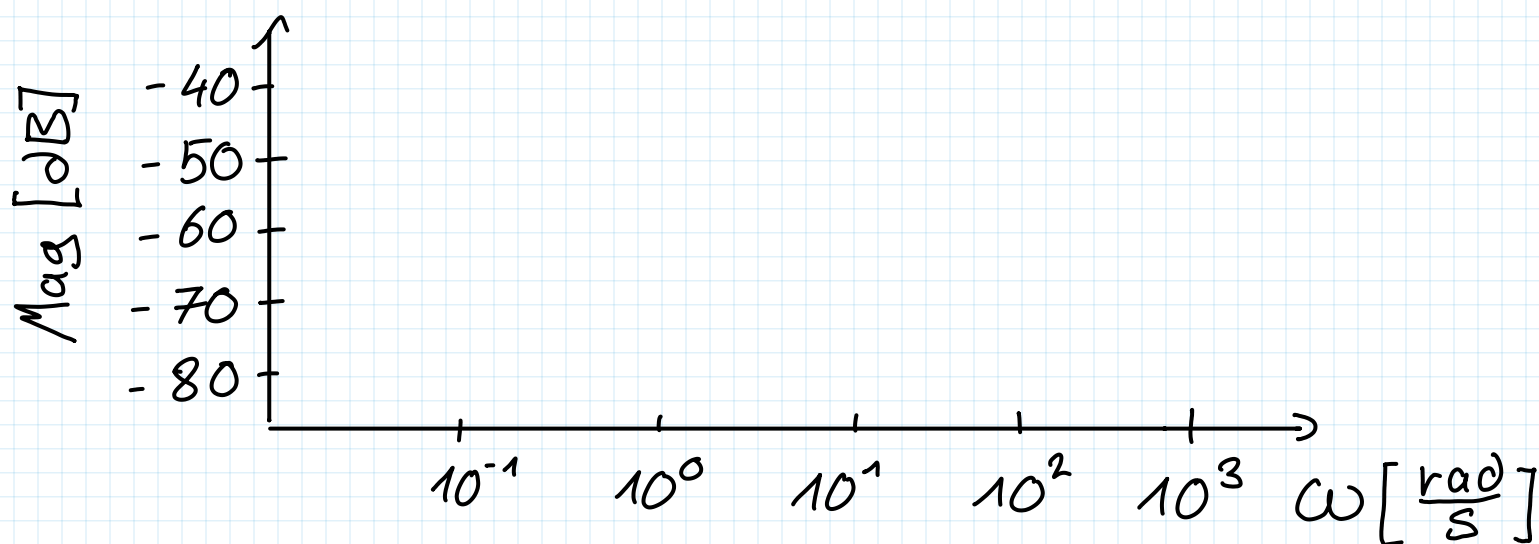
$$|G(0)|[\text{dB}] = 20 \cdot \log_{10}(|G(0)|) \approx$$

$$Z_1 =$$

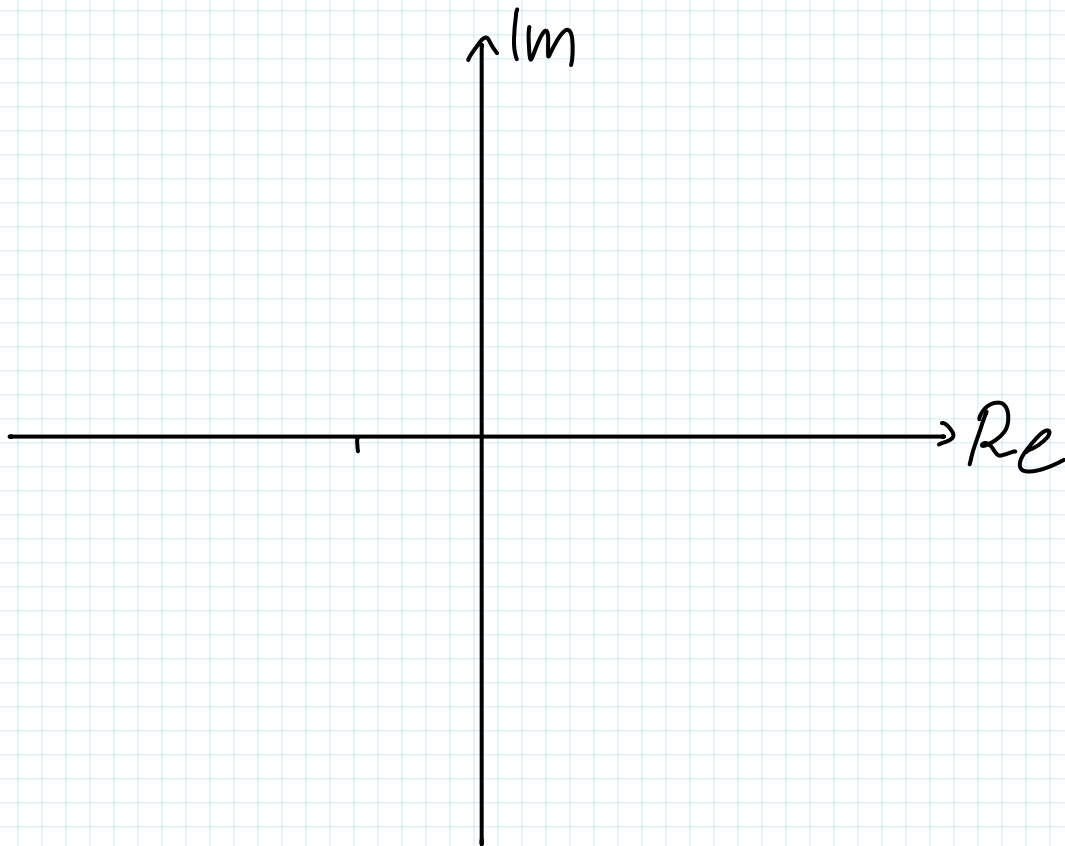
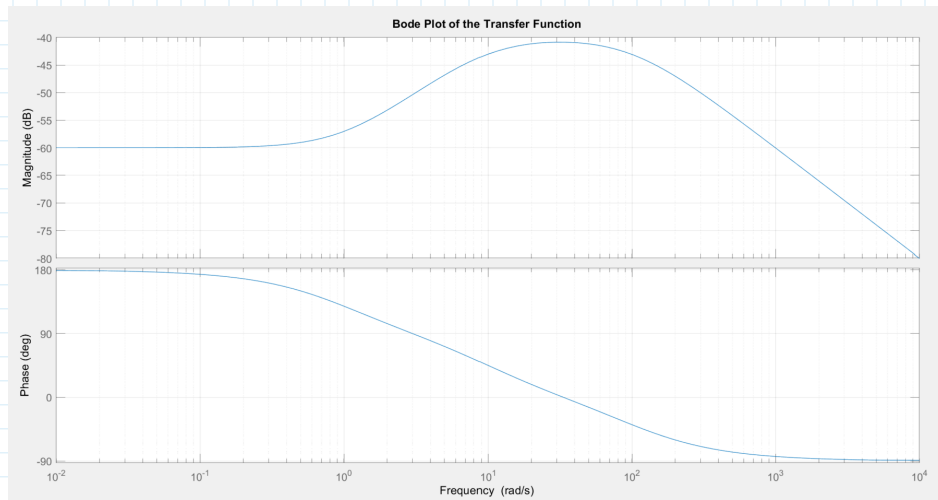
$$P_1 =$$

$$, P_2 =$$

Magnitude		-20 dB/dec	+20 dB/dec
Phase	-90°	stable pole	non-minimum phase zero
	+90°	unstable pole	minimum phase zero



• Now we extract the polar plot



These two plots show the same thing!
=> Will be very important for stability analysis.

Q1) A system with a double integrator will have the following slope of the magnitude plot at $\omega(0)$:

☐ 20dB/dec

☐ -20dB/dec

☐ 40dB/dec

☐ -40dB/dec

Q2) Assume you have the Bode Plot of $G(s)$. What changes if you plot $G^{-1}(s)$?

Q3)

Problem: Consider the transfer functions,

$$G_1(s) = \frac{s+0.1}{s^2}, \quad G_2(s) = \frac{s+1}{(s-0.1)(s+10)}, \quad G_3(s) = \frac{1}{s^2 + 0.02s + 1}, \quad G_4(s) = \frac{s+1}{(s+0.1)(s+10)}.$$

In Figure 13 the Bode plots of the aforementioned transfer functions are shown in a randomized order.

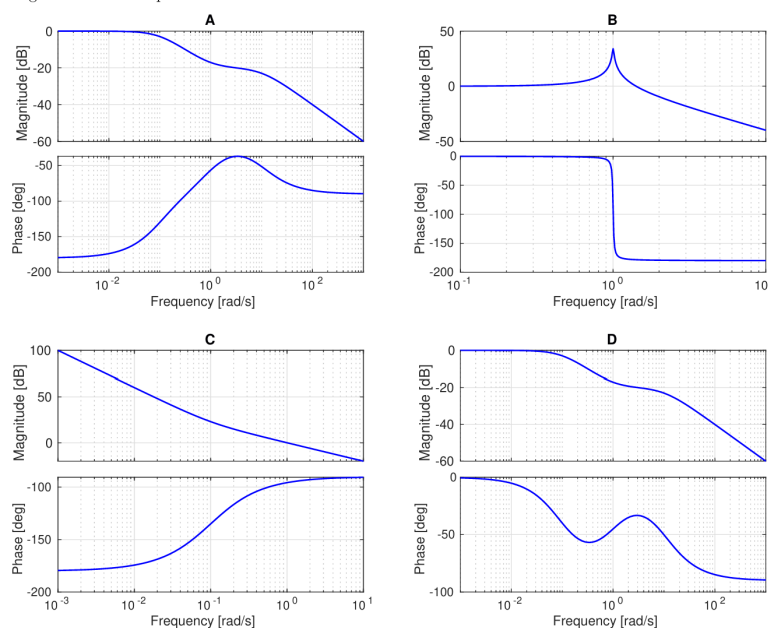


Figure 13: Bode plots of G_1 , G_2 , G_3 and G_4 in random order.

Q27 (1 Points) Mark the correct answer.

Mark the correct Bode plot and transfer function pairings?

☐ A $(A, G_1), (B, G_3), (C, G_2), (D, G_4)$

☐ B $(A, G_4), (B, G_3), (C, G_1), (D, G_2)$

☐ C $(A, G_2), (B, G_3), (C, G_1), (D, G_4)$

☐ D $(A, G_3), (B, G_2), (C, G_4), (D, G_1)$